

GLASGOW BISHOPTON, United Kingdom

WMO: 031340

Lat: 55.9067N

Lon: 4.5325W

Elev: 194

StdP: 14.59

Time Zone: 0.00 (EUW)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 24.4	(c) 27.4	(d) 20.7	(e) 15.7	(f) 28.4	(g) 23.2	(h) 17.6	(i) 30.4	(j) 32.2	(k) 44.8	(l) 28.6	(m) 46.5	(n) 3.6	(o) 330	(p) 0.628

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 13.9	(c) 73.3	(d) 63.1	(e) 69.9	(f) 61.1	(g) 67.1	(h) 59.7	(i) 64.5	(j) 70.8	(k) 62.7	(l) 67.9	(m) 61.1	(n) 65.5	(o) 8.3	(p) 330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.1	83.9	66.8	60.6	79.5	65.0	59.1	75.4	63.1	29.7	71.0	28.4	68.0	27.2	65.5	72.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 23.8	(b) 21.0	(c) 18.7	(e) 21.4	(f) 79.9	(g) 3.8	(h) 3.8	(i) 18.7	(j) 82.6	(k) 16.4	(l) 84.9	(m) 14.3	(n) 87.0	(o) 11.6	(p) 89.8		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			20.7	67.9	3.7	1.8	18.0	69.2	15.9	70.3	13.8	71.3	11.1	72.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.5	39.4	39.9	42.2	46.8	51.3	56.4	59.5	58.2	54.8	49.3	43.4	39.9		
	DBStd	8.61	5.36	4.86	4.73	4.27	4.74	3.90	3.80	3.33	3.88	4.40	5.48	6.60		
	HDD50	1600	330	282	243	113	39	1	0	0	7	65	205	315		
	HDD65	6047	795	702	706	546	426	260	178	214	306	486	649	779		
	CDD50	1040	0	1	3	17	78	192	294	253	151	44	6	1		
	CDD65	10	0	0	0	0	0	1	7	2	0	0	0	0		
	CDH74	89	0	0	0	0	6	21	50	11	1	0	0	0		
	CDH80	8	0	0	0	0	0	4	3	1	0	0	0	0		
(14) Wind	WSAvg	8.5	9.5	9.1	9.6	9.1	8.5	7.7	7.4	7.8	8.1	8.2	8.5	8.8		
(15) Precipitation	PrecAvg	46.1	5.1	3.7	4.0	2.5	2.8	2.6	2.7	3.5	4.6	5.1	4.9	4.9		
	PrecMax	60.4	10.9	9.8	8.6	5.1	6.5	4.6	6.5	7.3	11.3	11.0	8.1	10.9		
	PrecMin	34.7	0.7	0.4	0.8	0.2	0.3	0.5	0.5	0.4	0.4	1.3	0.7	1.2		
	PrecStd	6.1	2.4	2.2	1.9	1.1	1.5	1.1	1.3	1.7	2.2	2.5	1.9	2.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.6	53.1	58.3	66.1	74.1	76.5	79.5	74.6	69.2	61.8	57.3	54.3		
		MCWB	50.8	50.0	50.4	55.4	61.9	64.0	66.6	65.2	60.9	57.7	55.2	52.6		
	2%	DB	50.4	50.8	53.4	60.7	68.2	71.3	74.1	69.6	65.6	59.4	54.8	52.3		
		MCWB	48.8	48.6	48.7	52.1	58.3	61.1	64.5	61.8	59.2	56.1	52.7	50.8		
	5%	DB	49.0	49.1	51.2	56.9	63.4	67.4	70.3	66.9	63.1	57.6	52.9	50.8		
		MCWB	47.4	47.0	47.5	50.0	55.2	59.0	62.0	60.2	58.1	54.9	50.9	49.2		
	10%	DB	47.4	47.6	49.4	54.3	59.7	64.3	67.1	64.6	61.1	55.9	51.2	49.1		
		MCWB	45.7	45.5	46.1	49.1	52.8	57.5	60.3	59.0	57.0	53.4	49.4	47.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.3	51.2	52.3	55.9	63.4	65.1	68.1	67.0	63.6	59.1	55.6	53.0		
		MCDB	52.4	52.3	56.1	63.0	72.7	74.5	77.4	72.1	67.1	60.8	56.7	53.9		
	2%	WB	49.3	49.0	50.1	53.4	59.2	62.3	65.3	63.3	60.8	57.3	53.2	51.2		
		MCDB	50.1	50.5	52.5	58.4	66.1	69.2	72.4	67.1	63.3	58.6	54.5	52.1		
	5%	WB	47.7	47.5	48.3	51.3	56.4	60.0	63.2	61.6	59.3	55.6	51.3	49.5		
		MCDB	48.8	49.1	50.5	55.9	62.0	65.9	68.7	65.3	61.9	57.1	52.5	50.7		
	10%	WB	45.9	45.7	46.6	49.6	54.2	58.2	61.4	60.1	57.9	54.0	49.5	47.6		
		MCDB	47.4	47.4	49.2	53.7	58.6	63.1	66.0	63.6	60.4	55.6	51.0	49.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.0	9.6	11.2	13.6	14.8	14.3	13.9	13.2	12.0	10.9	9.4	8.8		
		MCDBR	9.0	9.6	13.1	18.1	21.1	20.3	19.3	16.6	14.4	11.8	9.6	9.7		
	5% WB	MCWBR	8.9	7.9	9.2	11.0	12.4	11.9	10.6	10.1	9.2	9.0	9.0	9.5		
		MCWBR	9.0	8.9	11.3	16.1	18.5	18.1	17.0	14.6	11.8	10.2	9.7	9.7		
(40) Clear-Sky Solar Irradiance	taub		0.330	0.342	0.366	0.383	0.386	0.382	0.383	0.367	0.359	0.342	0.321	0.315		
	taud		2.208	2.312	2.289	2.281	2.318	2.363	2.365	2.420	2.445	2.422	2.300	2.184		
	Ebn at Noon		175	227	251	265	271	273	270	268	253	226	180	148		
	Edh at Noon		19	24	31	36	37	36	35	31	27	22	17	15		
(44) All-Sky Solar Radiation	RadAvg		146	356	703	1150	1484	1452	1390	1117	774	416	194	110		
	RadStd		21	44	56	91	138	159	140	89	61	57	19	14		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page